

OptiCrop: Data Flow Diagram

Level 0 (Context)

Farmer (User) → provides soil/climate inputs → Web Form → Flask Backend → Trained ML Model → Predicted Crop → displayed back to Farmer.

Level 1 (Detailed)

- 1. User submits N, P, K, temperature, humidity, pH, rainfall via the web form (index.html).
- 2. Flask /predict route receives form data and structures it into a DataFrame.
- 3. Data is transformed using the saved StandardScaler (scaler.pkl).
- 4. Scaled data is passed to the trained Logistic Regression model (model.pkl).
- 5. The numeric prediction is decoded back into a crop name using the saved LabelEncoder (label_encoder.pkl).
- 6. The predicted crop name is rendered on result.html and returned to the user.

Offline / Training-time Data Flow

- Crop_recommendation.csv is loaded into the notebook for EDA and cleaning.
- Features are scaled and the label is encoded; data is split into train/test sets.
- Logistic Regression is trained and evaluated; K-Means is separately run for clustering insight.
- model.pkl, scaler.pkl, and label_encoder.pkl are serialized for use by the Flask app.